Hacettepe University / Computer Engineering
CMP 720 EMBEDDED SYSTEMS DESIGN

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Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

This is me and my partner's project.

Group 2

Yusuf Çağrı Öğüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

This project addresses remote agriculture monitoring and control in off-grid hobby gardens where internet access may be limited. This is important because irrigation systems need reliable, secure and low-power communication. Main idea here is to combine Cellular SMS for remote commands with LoRa for local sensor and actuator communication. I think the strengths are the realistic problem definition, low-power focus, and clear evaluation metrics such as latency, packet delivery ratio, energy consumption, and unauthorized command rejection. However, the related work could be more comprehensive, and the architecture part could include more technical specifications such as hardware interfaces, message format, and failure handling. I think I would also make the regulation and security requirements stricter, especially because SMS-based control may be vulnerable to replay attacks. I think the most critical part is reliable communication between the cellular gateway and LoRa nodes under low energy constraints. The evaluation plan looks mostly enough, but longer outdoor tests would improve it in my opinion.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

This project focuses on implementing DeFT, a deadlock-free and fault-tolerant routing algorithm for 2.5D chiplet networks. I found this topic very interesting and suitable for further research, because chiplet-based systems are becoming more important and routing must handle reliability, timing, and should also be scalable. The main idea is to use strict virtual network rules, lookup tables, and fault-aware vertical link selection to keep the network reachable and deadlock-free even when some vertical links fail. I think the project name and motivation are strong, and I would definitely like to see the continuation of this work. However, the presentation was a little too simple for such a complex topic, so it could be explained more clearly for people who are not familiar with 2.5D integration, and other notions in the context. More diagrams, especially a pipeline diagram and a clearer routing example, would make the methodology easier to follow. The most critical part seems to be correctly implementing the deadlock avoidance rules and fault injection. I believe the evaluation plan is good because it includes comparison with XY routing, but maybe it can be extended by also dealing with the LUT memory overhead?

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

This project focuses on real-time decoding of CAN bus traffic in vehicles, where raw CAN frames must be converted into meaningful information for diagnostics and monitoring. Since I do not have deep knowledge about CAN bus systems, I could not fully understand the exact role of DBC files and how they are used in this project, so this part could be explained more clearly. The main methodology as far as I understood is based on receiving CAN frames, buffering them, decoding selected signals, and outputting them through UART and SD logging with Zephyr RTOS support. I think the hardware architecture is one of the strongest parts of the project, especially the STM32, MCP2515, transceiver, UART, and SD card structure. However, the problem definition could be presented in a broader and clearer way, and the presentation did not feel fluent enough, maybe because it was not studied enough and took longer than expected. The evaluation plan includes useful scenarios such as high bus utilization, SPI contention, and burst traffic.

Group 5

Metin Irkıçatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*

Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

This project is about contention-aware task duplication for DAG scheduling on NoC-based multiprocessor systems. Main idea of the project is to extend a HEFT-like list scheduler with a duplication decision rule that considers contention-aware communication delay, instead of applying duplication only with an ideal communication model. I think the topic is technically valuable and close to real scheduling problems, but the presentation could have been more organized and better prepared, especially because scheduling, task duplication, and NoC contention are not easy concepts for everyone. The methodology is good, but I would improve the duplication rule by not only checking if EFT difference is positive, but also adding a threshold or cost function for memory overhead, link utilization, and duplication count. Otherwise, small EFT improvements might still create unnecessary complexity. The most critical component seems to be the contention model, since inaccurate link utilization or congestion estimation can directly lead to wrong duplication decisions. The evaluation plan includes useful metrics such as makespan, speedup, communication latency, network utilization, and duplication ratio, but they might also make it stronger with more detailed stress cases under high CCR workloads.

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

Group 9

Mehmethan Ayrım

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

This project's main focus is acoustic source localization for search and rescue operations after disasters, which is a very critical and important problem because traditional communication or visual methods may fail under rubble. I think the layout, problem/motivation part, requirements, constraints, and embedded system design considerations were very well prepared and presented well. The project clearly considers timing, synchronization, low bandwidth communication, energy efficiency and reliability. The main idea is to use distributed ESP32-S3 sensor nodes with microphones, GNSS PPS synchronization, LoRa communication, TinyML-based acoustic event filtering, and a Jetson Nano central hub for multilateration. I think this is a strong and meaningful system architecture. However, I could not fully understand the methodology part, especially the central hub localization steps such as Chan-Ho, Taylor Series, Levenberg-Marquardt, and Nelder-Mead. It looked a little complex and confusing for people who are not familiar with localization algorithms, including me. The step separation in the methodology was good, but each step could be explained with simpler intuition and more detail. I believe the most critical part is keeping microsecond-level synchronization between nodes, because small timing errors can directly damage TDOA localization accuracy. The evaluation plan looks strong, with localization accuracy, latency, synchronization drift, packet delivery ratio, and TinyML F1-score. I would like to see more explanation about how these tests will be physically performed in a rubble-like environment.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

This project focuses on a closed-loop Galvanic Vestibular Stimulation system for real-time balance assistance, which is a safety-critical and interesting embedded system problem. I think the problem definition is strong because it connects balance assistance with real-time control, sensor fusion, low latency and most importantly human safety. The main idea is to read IMU data, estimate posture with complementary filter and Kalman filtering, compute stimulation current with a PID controller, and enforce safety through hardware limits, RTOS monitoring, and watchdog protection. I especially liked the FreeRTOS task architecture, timing analysis, and multi layer safety design. The RMA-based schedulability analysis and explicit WCET values also make the proposal more convincing. One thing I might do differently is to add a small stress-test scenario for RTOS behavior, for example measuring jitter when logging load increases or when sensor data arrives with occasional delays. This would show whether the safety and control tasks are really isolated from lower-priority work. I think the most critical component is the safety path, because no software or timing fault should create unsafe stimulation current. I believe the evaluation plan is mostly sufficient with latency, jitter, sway reduction, tracking error, and fault injection tests, but in my honest opinion, maybe it could be improved by explaining how the simulated sway or pendulum setup represents real human balance behavior.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

This project is about GPU accelerated quadrotor simulation and controller evaluation on an NVIDIA Jetson. I think the problem definition was strong, because the presentation clearly showed why static controllers may fail under noise and environmental changes. The altitude response graph was also useful for understanding controller behavior. The main idea is to run the real-time quadrotor simulation and PID control on the CPU while the GPU evaluates many candidate PID gains in parallel using Monte Carlo simulations. I liked the monitoring PC part and the explanation of why they use simulation instead of a real drone, because using a real drone would increase cost, safety risk, and implementation complexity. The advantages and disadvantages of shared CPU-GPU memory were also given which was nice. However, I could not fully understand what exactly will be sent to the GPU and how the GPU result will update the controller in practice while the presentation was being presented. I believe CPU's role was explained more clearly than the GPU's role. The literature review could also be a bit more detailed, especially about trusted benchmarks, datasets, or reference trajectories used in this field. The evaluation metrics and hard constraints seemed a little short and random? I did not fully understand why limits such as 1000 simulations per second, 15 seconds update time, and 2 KB local memory were selected. I think the most critical part is synchronizing real-time CPU simulation with parallel GPU Monte Carlo evaluation without breaking timing constraints. I also think the evaluation plan should explain more clearly whether the final comparison will be based on response graphs, trajectory tracking error or runtime performance. As future work, if feasible, this system could be executed on a real drone.

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*

Group 17

Oğuz Kağan Yağlıoğlu, Emre Esef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

The project focuses on predictive maintenance of BLDC motors by using vibration, acoustic, temperature, and operating condition data to detect motor faults before they cause serious failures. I think this is a well chosen and important embedded system topic because it directly relates to low-power sensing, wireless communication, and real-time maintenance decisions. I do not have deep knowledge about BLDC motors, and I learned the basic idea of them during this presentation, so I don't think I can make very detailed technical comments about the motor side. However, the presentation was clear enough for me to understand the main goal and system flow. The proposed architecture, including the edge sensing node, LoRaWAN communication, fog-based ML processing, and condition-aware learning pipeline, seemed well thought and realistic for me. I especially liked that the study already produced meaningful results, such as feature selection, class merging, condition-aware modeling, and 92% file-level accuracy. The explanation of why some fault classes were merged into an "uncertain_fault" class was also reasonable, because it reflects real limitations. The strongest part of the work is that it combines hardware design, sensor selection, ML pipeline, and practical deployment considerations. As an improvement with future work, I would like to see more discussion about how the model will perform on real BLDC motor data, not only on public dataset results. The evaluation plan looks promising, but as I said, future tests on their own motor setup might be essential.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

This project focuses on multi application task mapping in many core NoC systems, which is an important problem because poor mapping can increase communication cost, latency, thermal hotspots, and reduce system efficiency. The topic was deep and interesting, but I think the problem definition and novelty could be explained more simply for people who are not familiar with NoC mapping or NSGA-II. At first, I did not fully understand the exact problem and contribution, but after our instructor asked the presenter to explain it again in Turkish, it became much clearer. The main methodology combines spectral clustering and pre-processing with NSGA-II, so that better initial elite seeds are generated and the search space is reduced before evolutionary optimization. I think the methodology part was stronger than the problem definition, because it was explained more clearly with stages such as graph consolidation, spectral partitioning, elite seeding, and evolutionary tuning. A more detailed topology figure could improve the presentation, especially for understanding the 9x9 NoC structure and mapping constraints. The most critical part seems to be designing good spectral seeds without reducing population diversity too much. The evaluation plan seems well enough with MATLAB/PlatEMO, NoC benchmarks, and baselines like random NSGA-II and static spectral mapping, but it could also include more explanation about runtime overhead?

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

This project's goal is to implement or emulate an autonomous obstacle avoidance vehicle using a LIDAR sensor and an embedded system. This is actually a very important embedded system problem because the system must process sensor data, make movement decisions, and control motors with very low delay, while also considering safety, reliability, resource limits, and energy consumption. The main idea is mostly taken from an existing paper, where a Raspberry Pi, RPLidar A2, motor driver, and DC motors are used for indoor navigation, and the proposal suggests emulating this system in Renode under different environmental conditions. I think the topic is interesting, but the scope does not look very realistic for one person unless it is strongly simplified, because modeling LIDAR data, vehicle movement, motors, GPIO behavior, and environment interaction in simulation can become a large project if you ask me. The presentation also seemed a bit messy, because the connection between the paper implementation and the Renode emulation plan is didn't sound very clear to me. The strongest part is that real time design considerations such as sensing to actuation delay, sensor noise, resource limits, and simulation versus real world limitations. However, I would modify the project by reducing the scope, defining a very clear minimum demo and explaining what will be implemented from scratch and what will only be simulated. The most critical component is probably creating a meaningful LIDAR/environment emulation and connecting it to the control algorithm. The evaluation plan didn't seem detailed enough either. I think it should include all these measurable metrics like obstacle avoidance success rate, reaction time, collision count, CPU usage, and behavior under noisy or missing sensor data.